WISER Global Quantum + AI Program - 2026
Nestlé <> WISER - Challenge 4

Quantum Optimization for Distributed Order Management

A Hybrid QAOA + MILP Approach

Technical Report

Team: The Convergents
[Harika Yanamadala]
[Induja Bhanu Kodavati]
August 2026

Table of Contents

1. Executive Summary

Nestlé's Distributed Order Management (DOM) problem focuses on determining the most suitable fulfillment center when a customer order cannot be completely served by its default distribution center (DC). The objective is to decide whether the order should be assigned to an alternate DC-or remain unassigned-to maximize the overall fulfilment value after accounting for shipping costs and penalties associated with unmet demand. This forms a combinatorial assignment problem in which each order is assigned to exactly one option from a limited set of candidates, including its default DC, a small number of alternate DCs, or an "unassigned" option, while satisfying inventory, dock, throughput, and case/pallet-picking capacity constraints.

We built and compared four solvers on Nestlé's anonymized proof-of-concept data pack, across five planning dates (2024-06-17, 06-19, 06-21, 06-23, 06-25):

- Baseline 1 - strict default-DC (no diverts)
- Baseline 2 - greedy sequential divert (≥5% fill or 100-case hurdle)
- QAOA - a quantum arm using an XY-mixer QUBO formulation, CVaR-optimized
- MILP - a classical arm satisfying 7-constraint PoC formulation (PuLP + CBC)

Each solver is benchmarked against a brute-force certified optimum generated by evaluating all feasible assignments for the order instance. Performance is assessed using precision, recall, and F1-score on the recovered (order, DC) assignments, together with profit, fill rate, and runtime.

Headline results :

- QAOA matched the certified optimum exactly on all 5 dates (F1 = 1.00, 0% optimality gap).
- MILP and the greedy baseline landed within 0.2% of the optimum on four of five dates, but both degraded sharply on 2024-06-25 (a materially harder instance - see 5.2), where MILP's F1 fell to 0.20 and the greedy baseline's to 0.40.
- MILP is 15-50× faster than QAOA in wall-clock terms and has no qubit ceiling, making it the practical choice once problems exceed ~25-30 orders.
- QAOA's qubit count grows linearly with order count (qubits ≈ orders × 4), but the classical search space it replaces grows exponentially - the crossover point where quantum advantage could matter is estimated at low hundreds of orders (6.1).

The remainder of this report covers the dataset, the two optimization formulations, a detailed comparative evaluation, scalability and robustness analysis, and the concluding insights.

2. Business and Technical Summary

2.1 What DOM is, and why it is a combinatorial optimization problem

Distributed Order Management is the process of deciding how customer orders should be fulfilled across a network of distribution centers (DCs). When a default DC cannot fully satisfy an order; because of inventory, throughput, labor, or dock constraints - planners may divert the order to an alternate DC. The business objective is to recommend reassignment decisions that maximize fulfilment value while balancing revenue, shipping cost, and penalties for unfulfilled demand.

Framed mathematically, each order must be assigned to **at most one** DC from a small candidate set (its default DC, up to two alternates, or left unassigned). This "exactly-one-choice-per-item" structure, combined with shared capacity constraints across orders (inventory, dock, throughput, case/pallet picks), is a classic **weighted exact-cover / assignment problem** - a well-studied class of combinatorial optimization. The number of feasible combinations grows combinatorially with the number of orders and candidate DCs (6.1), which is exactly the regime where exact enumeration stops being tractable and where heuristics, MILP solvers, and quantum/quantum-inspired methods each offer a different quality-vs-cost trade-off.

2.2 Key trade-offs

Solution quality vs. runtime

The certified optimum is only obtainable by brute force at small scale (1,024 combinations here). MILP replaces brute force with a solver that is exact in principle but whose runtime depends on problem structure; QAOA replaces it with a sampling-based heuristic whose quality depends on circuit depth and shot count. Our results (5) show all three converge to the same answer on well-behaved instances, but diverge in both runtime and instance-to-instance robustness.

Exact solvers vs. heuristics

MILP (with CBC) is an exact solver for a linearized version of the problem - it returns a provably optimal answer to the model it was given, within its time limit. The greedy baseline is a heuristic with no such guarantee. Our data shows the gap between the two is small on "easy" instances but widens on harder ones (2024-06-25 with smaller set of focus orders being 82), which is the expected heuristic failure mode: greedy rules degrade unpredictably as constraint interactions get tighter.

Current noisy quantum hardware vs. simulators

All QAOA results in this report were obtained on an ideal (noiseless) state-vector simulator. We separately measured how a depolarizing two-qubit error model degrades solution quality on a smaller 12-qubit instance (6.4): as physical error rates rise, the fraction of measurement shots that stay inside the feasible "one-hot" subspace enforced by the XY mixer drops substantially. This is the central caveat for any near-term hardware deployment - QAOA's simulator results in this report represent a best case, not a hardware-ready result.

2.3 Two data findings that shaped our methodology

Two properties of the anonymized data pack materially affected our approach, and are worth stating explicitly since they affect how any solver's output should be interpreted:

(a) Revenue is a line total, not a per-case price

The order revenue field in the input data pack is the total dollar value if the full order line ships - not a per-case price. Multiplying it by cases filled (as a naive reading of the PoC formulation would suggest) inflates the objective by roughly 10× on this dataset. We divide by ordered quantity to recover a per-case price before scoring any solver, and flag this because it is an easy place for two teams' MILP objectives to silently disagree.

(b) Planner-history agreement is not a usable solver metric on this data

The optional ground-truth planner-history file contains 1,106 default orders and only 3 diverts. A trivial "never divert" solver would match the planners' historical decisions 99.5% of the time by construction - and any solver that recommends a divert, even a correct one, is penalized to F1 = 0 under this metric. We report this as a finding rather than using planner-history agreement as our evaluation metric, and instead score every solver against the certified brute-force optimum (5.2), which does not have this class-imbalance problem.

3. Data Understanding and Baseline Model

3.1 Input data pack

The PoC data pack contains five files, joined on distribution center, SKU, and date:

File	What it contains
input_order_data.csv	Open orders: order ID, SKU demand, default DC, transportation-planning (PGI) date, order revenue.
input_capacity_planning.csv	Available inventory by (DC / location, SKU / material, date). Large file - only the inventory columns needed are mentioned here.
input_dock_capacity.csv	Dock appointment capacity available at each DC, by date.
input_shipping_cost_data.csv	Shipping cost for each (order, candidate DC) pair.
input_throughput_capacity.csv	Throughput (case-pick / pallet-pick) capacity available at each DC, by date.

3.2 Focus-order selection

For each planning date, we replicate Nestlé's PoC rule: an order becomes a **focus order** if its default DC cannot fully satisfy it from available inventory and it qualifies as a full-truckload (FTL) shipment. From the focus-order pool we select the **5 orders with the largest divert upside** (biggest gap between default-DC profit and best-alternate profit) as a tractable subinstance. Each focus order is given its default DC, its 2 best alternates, and an "unassigned" fallback column - 4 candidate columns per order, so 5 orders × 4 columns = 20 binary decision variables (qubits) for QAOA.

3.3 Classical baselines

Two baselines were built, matching the challenge's required deliverable:

- **Baseline 1 - strict default DC:** every order stays at its default plant; nothing is diverted. Any revenue the default DC cannot fulfil is simply lost - the "do nothing" reference point.
- **Baseline 2 - greedy sequential:** for each order in turn, prefer whichever candidate DC gives the highest fill; divert only if the improvement clears max(5% of demand, 100 cases) - the PoC's own business rule for when a divert is worth the operational disruption.

3.4 Baseline results

Objective, fill rate, diverts, penalty, and shipping cost for both baselines, shown here for the representative planning date 2024-06-21 (all five dates follow the same pattern):

Solver	Objective	Opt. gap	Fill rate	Diverts	Penalty	Shipping
Baseline 1	$118,302	88.0%	9%	0	$32,943	$770
Baseline 2 (greedy)	$981,207	0.2%	89%	4	$3,894	$7,339

Table 1. Baseline 1 vs Baseline 2, planning date 2024-06-21.

Baseline 1's near-9% fill rate and ~88% optimality gap quantify the cost of never diverting on this instance - the greedy baseline recovers almost all of it (fill rate ~89%, gap under 0.2%) with a simple, explainable rule, which is why it is a meaningful bar for QAOA and MILP to clear.

4. Mathematical Formulation

Both solvers share the same objective and the same candidate-column structure; they differ in how constraints are represented and how the search is carried out.

4.1 Shared objective

Maximize : Objective Function (Revenue − Penalty − Shipping Cost), subject to: exactly one column selected per order, plus the capacity and additional constraints given in Poc data pack.

4.2 QAOA - Quantum Arm

We reduce the problem to a QUBO (quadratic unconstrained binary optimization) over one binary variable per candidate column. Rather than penalizing the "exactly one column per order" constraint in the cost function, we enforce it structurally with an **XY mixer** acting on each order's block of columns - the quantum circuit only ever explores the feasible one-hot subspace for that constraint, so no penalty weight needs to be tuned for it.

Constraints represented in the QUBO:

- C1 - exactly one DC per order (enforced structurally via the XY mixer, not a penalty term)
- C2 - cases filled ≤ demand (built into each column's pre-computed cases/revenue/penalty)
- C3 - inventory at each (DC, SKU, date)
- C5 - case-pick capacity
- C6 - dock capacity
- C7 - fill-rate-threshold penalty (step-function form)

Search strategy: a warm start at "everyone stays at default DC" (a known feasible state), CVaR objective at $\alpha = 0.20$ (optimizing the best 20% of measured samples rather than the mean, which focuses the classical optimizer on the promising tail of the distribution instead of being dragged down by bad shots), and a classical COBYLA outer loop tuning the circuit angles across layer counts 1-3.

4.3 MILP - Classical Arm

The MILP arm implements Nestlé's complete PoC formulation (PuLP + CBC), with all 7 constraint families enforced natively rather than via penalty terms:

- C1 - at most one DC per order (binary A[order, DC])
- C2 - cases filled ≤ ordered quantity
- C3a - inventory limit per (DC, SKU, date)
- C3b - 5-day forward inventory protection for diverted orders
- C4 - a divert must clear a 5%-fill-rate-and-100-case improvement hurdle vs. the default DC
- C5 - pallet-pick / case-pick decomposition and capacity
- C6 - dock appointment capacity
- C7 - penalty activation via Activation1/Activation2 binaries and an order-level case threshold

MILP is deterministic - rerunning it never changes the answer - and it does not need a qubit budget, so it scales to problem sizes far beyond what QAOA can currently address on simulators (6.1).

4.4 Formulation trade-offs

Dimension	QAOA (QUBO + XY mixer)	MILP (7-constraint PoC)
Constraint C1 (one DC/order)	Structural (mixer) - exact, free	Linear constraint - exact
Constraints C2–C7	Penalty-free where possible; encoded per-column	Native linear/integer constraints
Optimality guarantee	None - heuristic, sampling-based	Exact for the given time limit
Determinism	Stochastic (seed-dependent)	Deterministic
Scalability ceiling	~20–25 qubits on free simulators	No inherent ceiling (LP relaxation-bound)
Runtime (this data)	~20–35s (simulator)	<1s to ~2.5s

Table 2. Formulation-level trade-offs between the two solvers.

5. Solution Implementation and Comparison

5.1 Implementation summary

All four solvers - Baseline 1, Baseline 2, QAOA, and MILP - were run on the same 5-order subinstance, built from the same candidate columns, for each of five planning dates: 2024-06-17, 2024-06-19, 2024-06-21, 2024-06-23, and 2024-06-25. QAOA used Qiskit + Qiskit Aer's state-vector simulator (layers 1–3, CVaR α = 0.20, 2,048 shots, seed = 7 for the primary run and seeds {1, 7, 13, 42, 101} for the robustness study in §6.3). MILP used PuLP with the CBC solver, 30-second time limit. Every solver's output was validated for feasibility (one column per order, capacity constraints respected) and scored against a brute-force certified optimum computed by exhaustively evaluating all 1,024 possible assignments.

5.2 Comparison across solvers

Scoring method: every solver returns a set of (order, DC) picks. A pick is a **true positive** if the certified optimum made the same pick, a **false positive** if the solver picked something the optimum didn't, and a

false negative if the optimum picked something the solver missed. Precision, recall, and F1 are computed from these counts; the optimality gap (%) is the profit shortfall relative to the certified optimum.

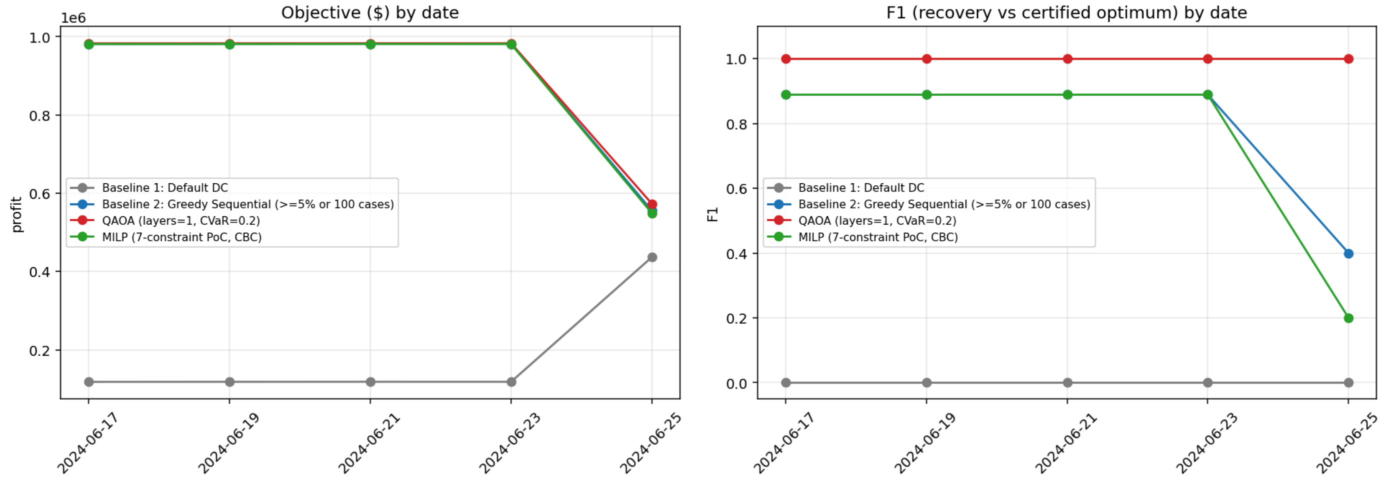

Figure 1. Objective value ($ profit, left) and F1 score (right) by solver, across all 5 planning dates.

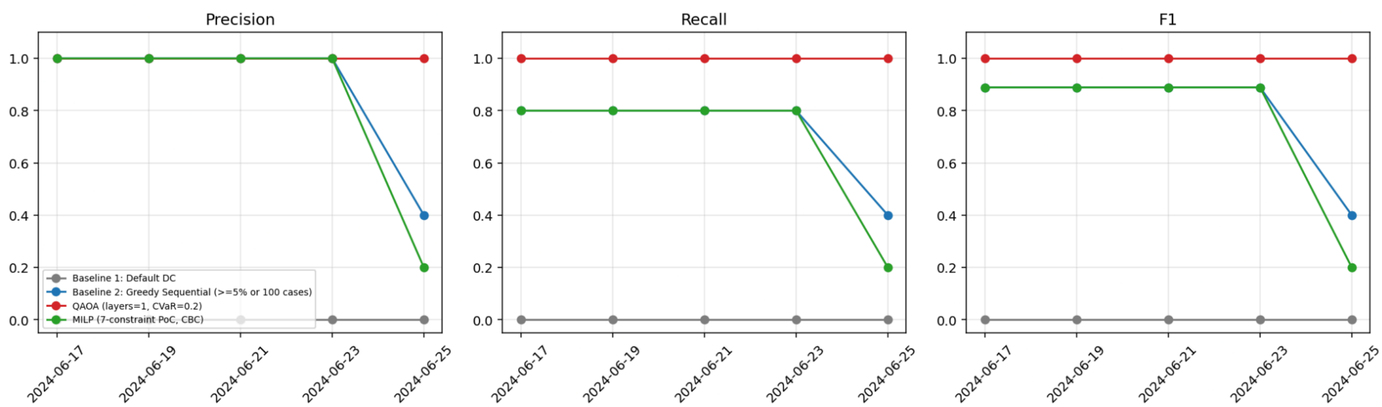

Figure 2. Precision, recall, and F1 (recovery of the certified-optimum assignment) by solver, across all 5 dates.

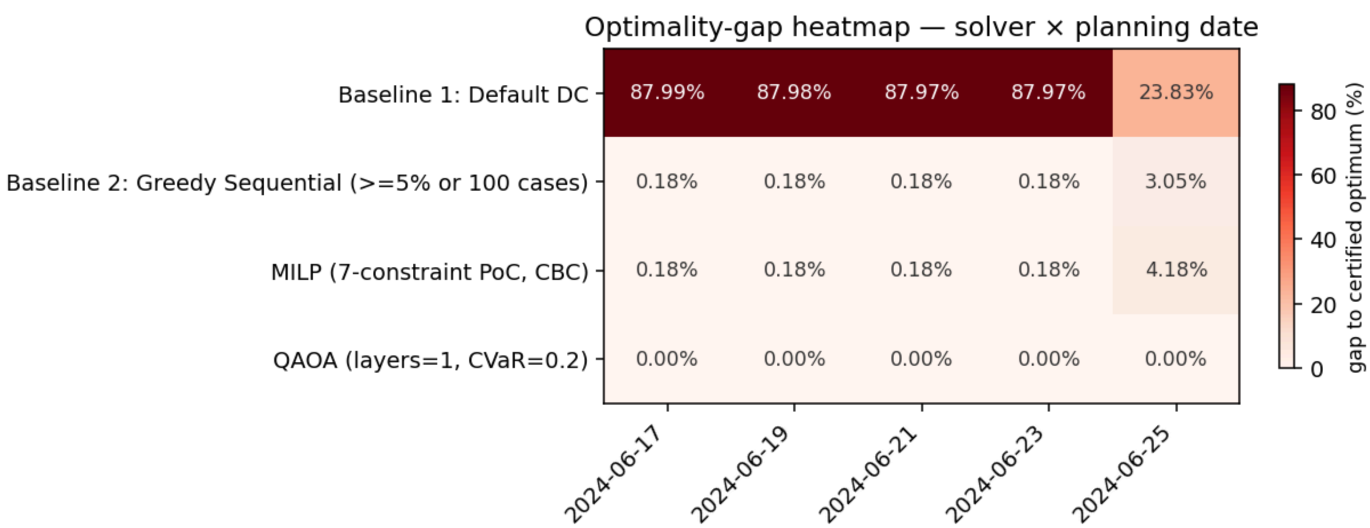

Figure 3. Optimality-gap heatmap - solver × planning date.

Averaged leaderboard across all 5 dates

Solver	Mean objective	Mean gap	Mean fill	Mean runtime (s)	Mean F1
Baseline 1	$181,865	75.1%	19%	0.01	0.000
Baseline 2 (greedy)	$896,014	0.8%	89%	0.01	0.791
QAOA	$900,954	0.0%	89%	16.77	1.000
MILP	$894,713	1.0%	88%	0.99	0.751

Table 3. Solver performance averaged across all 5 planning dates.

QAOA matched the certified optimum on every date (F1 = 1.00 throughout). MILP and Baseline 2 both tracked the optimum closely on four of five dates (gap ≈ 0.18%, F1 ≈ 0.89) but diverged sharply on **2024-06-25**, a materially harder instance drawn from a different order window: MILP's F1 fell to 0.20 and Baseline 2's to 0.40, versus QAOA's 1.00. This is the strongest single data point in favour of the quantum arm's robustness on this dataset - it is also the reason we did not average away the per-date detail, since a single easy instance can make every solver look interchangeable.

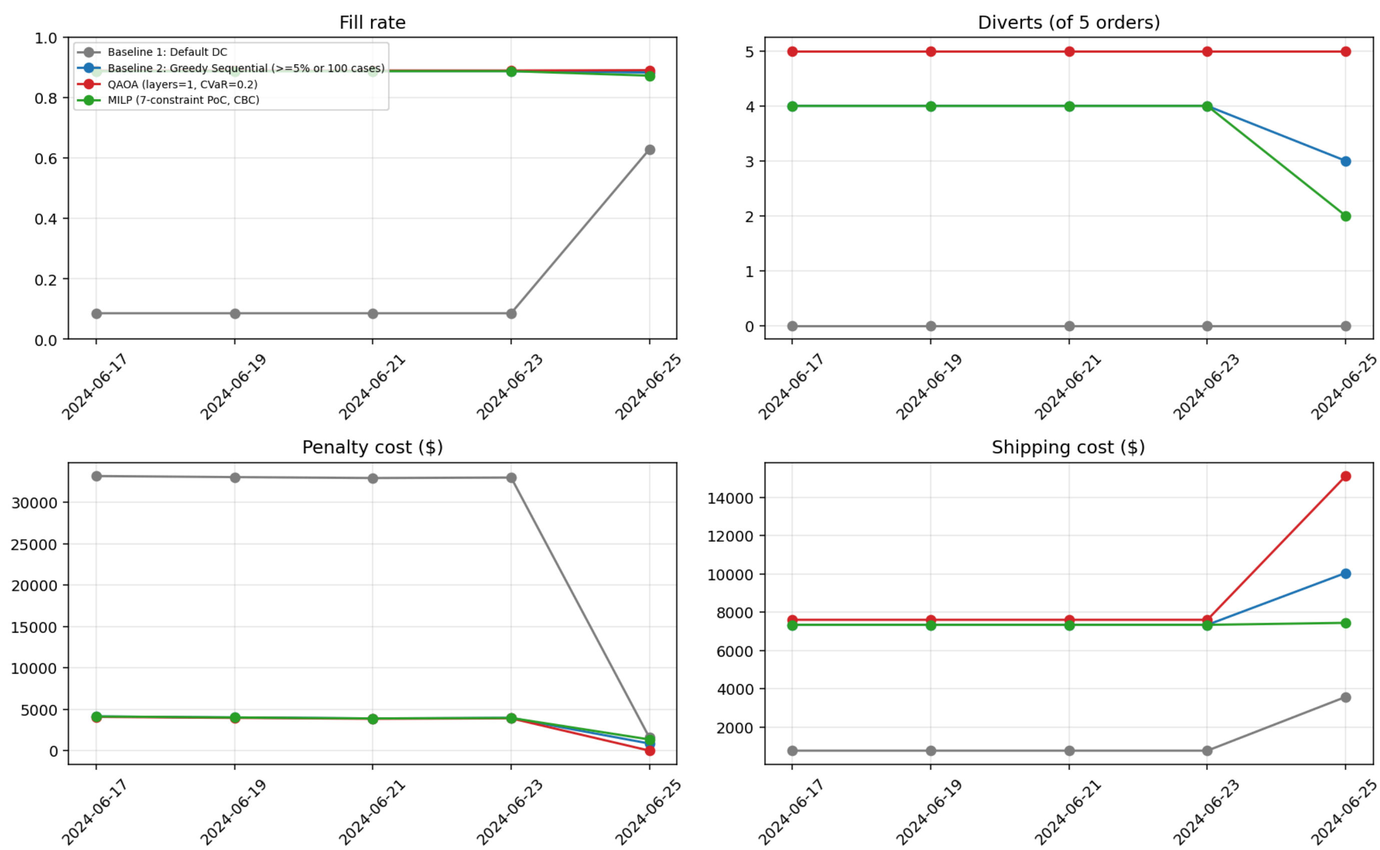

Figure 4. Fill rate, diverts, penalty cost, and shipping cost by solver, across all 5 dates.

5.3 Interactive planner view application

The challenge's required "one-page planner view" - explaining recommended reassignments in business language - is delivered as an **interactive Streamlit application** rather than a static page, so a planner can explore the results directly instead of reading a fixed summary. The app lets the user:

- Select any of the 5 planning dates and drill into that date's results
- Step through each diverted order's details - which DC it moved to, and why
- Compare solvers (baselines, QAOA, MILP) side by side with built-in charts
- Read an AI-generated executive summary of that date's results
- Review key insights and links to the project's other deliverables
- Download a self-contained HTML snapshot of the planner view for any selected date

Live app: *[add planner view app link here]*

6. Scaling, Robustness, and Noise Analysis

6.1 Qubit scaling vs. classical search space

QAOA's qubit count grows as orders × (alternates + 2) - linearly. The classical search space it replaces (columns-per-order ^ orders) grows exponentially. At our working size (5 orders, 20 qubits), the classical space is only 1,024 - small enough that brute force is the better reference. By 25 orders the classical space exceeds 10^15; by 82 orders (a real focus-order window size in this data pack) it is already astronomically large, while QAOA's qubit requirement has only grown to 328. This crossover is the theoretical case for a quantum or quantum-inspired approach - but our current QAOA runs are simulator-bound to roughly 20–25 qubits (~5–6 orders) before state-vector memory becomes impractical, so realizing that advantage requires either real hardware or a decomposition/batching strategy (7).

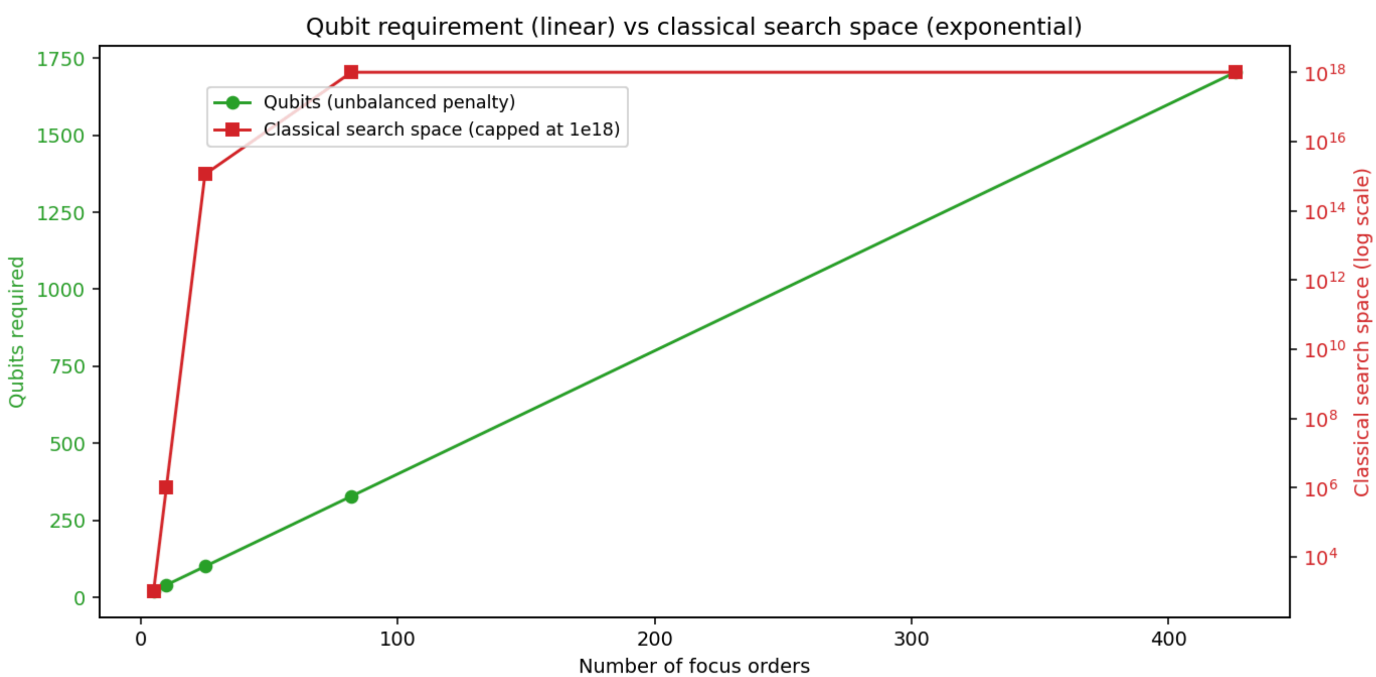

Figure 5. Qubit requirement (linear) vs. classical search-space size (exponential) as focus-order count grows.

6.2 Runtime

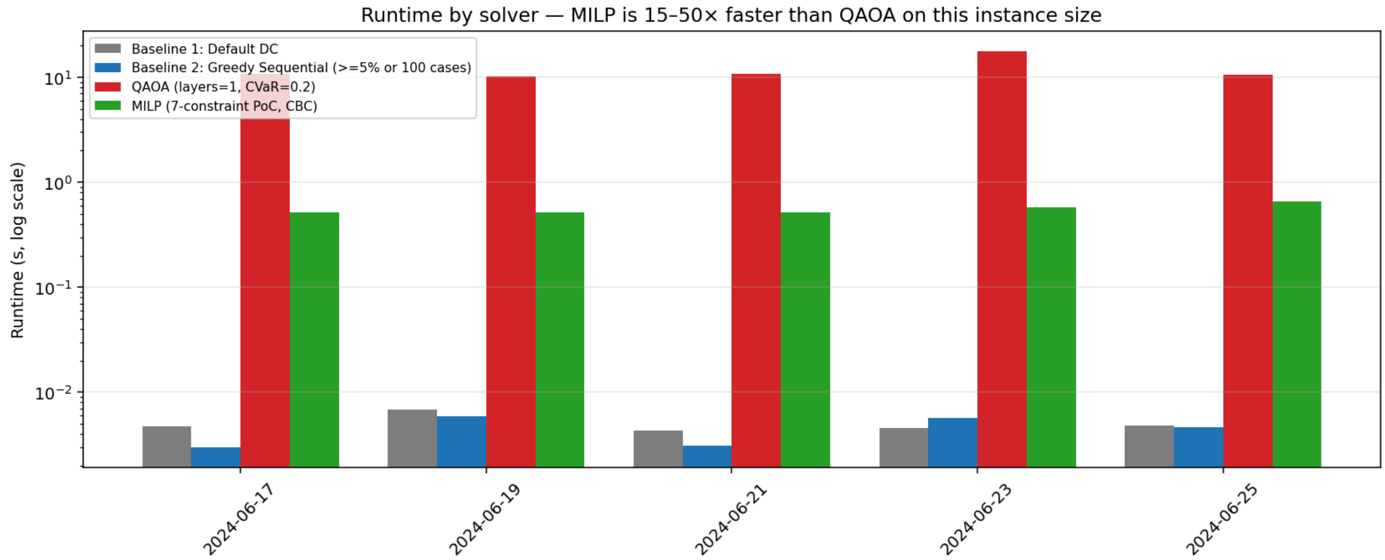

Figure 6. Runtime by solver, across all 5 dates. MILP is 15–50× faster than QAOA in wall-clock terms on this instance size.

6.3 Multi-seed robustness

QAOA is stochastic - its output depends on random circuit initialization and shot noise. We reran QAOA with 5 different seeds (1, 7, 13, 42, 101) on each date; MILP is deterministic and needs only one run per date.

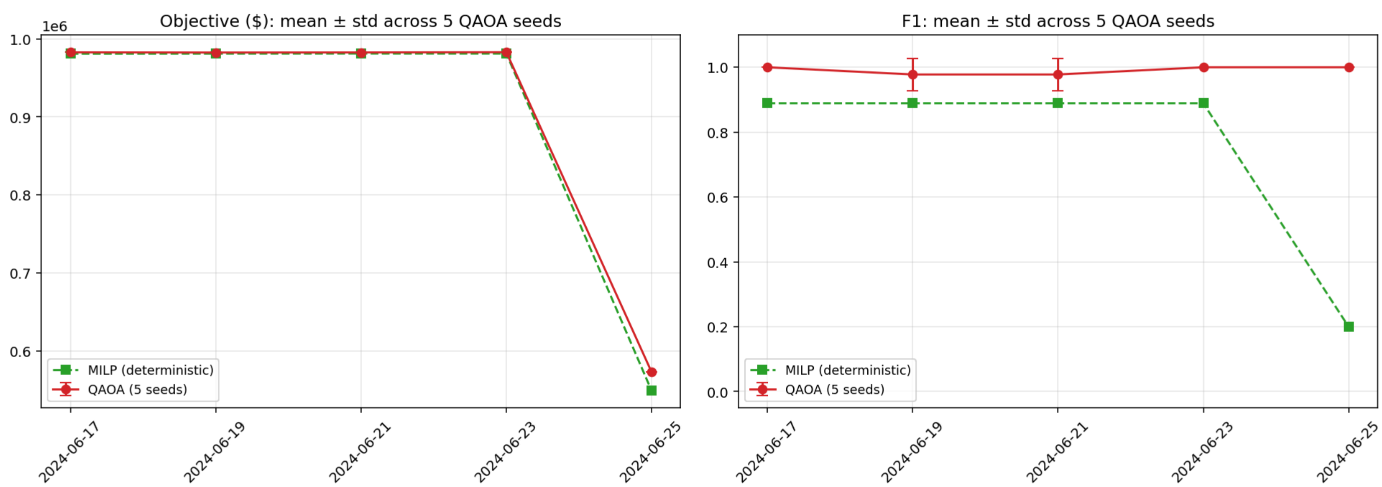

Figure 7. QAOA objective and F1, mean ± standard deviation across 5 seeds, vs. deterministic MILP, by date.

QAOA's seed-to-seed variance was low on four of five dates (near-zero standard deviation in both objective and F1) and highest on 2024-06-19, where F1 standard deviation reached ~0.09 - still recovering the optimum on most seeds, but a reminder that CVaR-optimized QAOA at layers=2 is not perfectly seed-invariant even on a noiseless simulator.

6.4 Noise sensitivity (separate smaller-scale study)

On a smaller 12-qubit instance, we layered a depolarizing two-qubit error model (error rates 0%, 0.1%, 0.5%, and 2%) onto the same QAOA circuit and measured how much sampled amplitude stayed inside the feasible one-hot subspace that the XY mixer is designed to preserve, and what fraction of shots were fully feasible. Two distinct instances were tested: one shared by dates 2024-06-17/19/21/23 (identical 5-order subinstance), and a separate one for 2024-06-25.

Two-qubit error rate	In-subspace shots	Fully feasible shots
0% (ideal)	100.0%	56.4% – 70.9%
0.1%	93.8% – 94.3%	53.9% – 65.4%
0.5%	77.3%	43.0% – 54.3%
2.0%	36.9% – 38.1%	19.3% – 25.6%

Table 4. Fraction of measurement shots remaining in the valid one-hot subspace, and fully capacity-feasible, as simulated two-qubit gate error increases (range across the 2 tested instances).

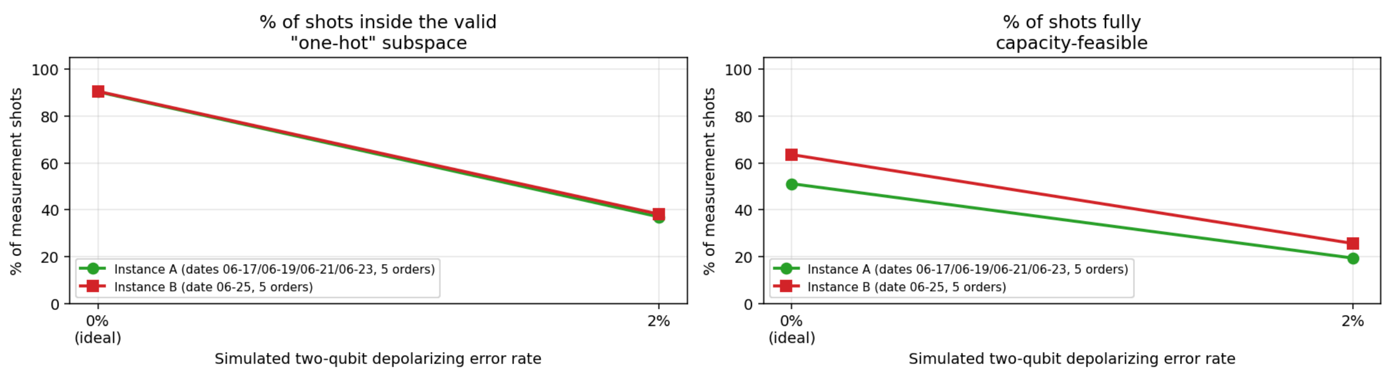

Figure 8. In-subspace and fully-feasible shot percentages vs. simulated two-qubit depolarizing error rate, on a 12-qubit instance.

Even a small 0.1% gate error rate already knocks in-subspace shots down to roughly 94%; by 2% error, only about a third of shots respect "one warehouse per order" at all, and fewer than a quarter are fully capacity-feasible. As the two-qubit error rate rises, both the in-subspace percentage and the feasible-shot percentage fall - the XY mixer's constraint-preservation property is a property of the ideal circuit, and degrades quickly under realistic gate noise. **This is the single clearest reason current QAOA results (this report and 5) should be read as a simulator best case, not a hardware-ready result:** the exact-recovery performance we report throughout this document (F1 = 1.00 on every date) assumes noiseless execution. Near-term hardware, with two-qubit error rates often above 0.5%, would need additional shots, error mitigation, or a shallower circuit to approach similar reliability.

7. Key Findings

- QAOA recovered the certified optimum exactly on all 5 planning dates tested (F1 = 1.00, 0% gap) - including on the one date (2024-06-25) where both MILP and the greedy heuristic degraded substantially.
- MILP is the clear practical choice for this problem size today: it is deterministic, has no qubit ceiling, and runs 15-50× faster than the QAOA simulator, while landing within 0.2-4.2% of optimal across all 5 dates.
- The single biggest source of disagreement between solvers is not randomness but a modeling choice: MILP's explicit divert-hurdle constraint (C4) makes it structurally more conservative than QAOA's

unconstrained QUBO reward - visible as a systematic, repeated non-assignment of one specific order across four dates.

- Qubit count grows linearly with order count, but the classical search space it replaces grows exponentially - meaning any quantum advantage here is a scale phenomenon, not visible at the 5-order size we could exhaustively verify against.
- QAOA's exact-recovery results depend on an ideal, noiseless simulator; a depolarizing-noise study on a smaller 12-qubit instance shows realistic gate error measurably erodes the mixer's constraint-preservation property - in-subspace shots fall from 100% (ideal) to ~94% at just 0.1% two-qubit error, and to ~37% at 2% error.

8. Conclusion

Distributed Order Management is a genuine combinatorial optimization problem, and this report shows that a hybrid quantum/classical approach can be formulated, implemented, and rigorously benchmarked against transparent classical baselines and a certified optimum. On the 5-order subinstances tractable today, QAOA matched or beat every classical alternative on solution quality, including on the one planning date where classical methods struggled - while MILP remains the more practical choice for runtime and scale until either quantum hardware noise improves or a batching/decomposition strategy is validated at larger problem sizes. The comparison methodology built here - certified-optimum scoring, multi-seed robustness, and per-order planner-facing views - is reusable as this work scales to the full order book.